PROJECT REPORT

TASK 25: BASIC DATA CLEANING USING PANDAS

1. Abstract

This project demonstrates a basic data-cleaning workflow using Python and Pandas. It combines three important operations: missing-value handling, duplicate-record removal, and data-type correction.

A manually created student performance dataset is used to identify data-quality issues, apply appropriate cleaning techniques, and produce a cleaned dataset.

The project highlights the importance of preparing reliable data before performing further analysis or machine-learning modelling.

2. Introduction

Data cleaning is an essential step in the data analysis process. Real-world datasets may contain missing values, duplicate records, and incorrectly formatted information.

These issues can reduce data quality and affect the accuracy of statistical analysis and machine-learning models.

Python's Pandas library provides useful functions for identifying and correcting these problems.

This project combines fundamental data-cleaning techniques into one practical workflow using Google Colab.

3. Objectives

The main objectives are:

- To inspect an original dataset.
- To identify and remove duplicate records.
- To identify missing values in different columns.
- To handle missing numerical and categorical values.
- To convert incorrect data types into appropriate formats.
- To compare original and cleaned datasets.
- To export the final cleaned dataset.

4. Tools and Technologies

Python: Used to develop and execute the data-cleaning workflow.

Pandas: Used for dataset creation, inspection, duplicate removal, missing-value handling, and data-type conversion.

NumPy: Used to represent missing values.

Google Colab: Used as the Jupyter Notebook development environment.

CSV: Used to store and export original and cleaned datasets.

5. Dataset Description

The project uses a manually created student performance dataset containing 12 original records and seven columns.

The columns are:

Column	Description
student_id	Unique student identifier
student_name	Student name
department	Academic department
age	Student age
marks	Marks obtained
attendance	Attendance percentage
joining_date	Joining date

The dataset intentionally includes two duplicate records, missing values in selected columns, and an incorrectly formatted age value.

All records are fictional and used for educational demonstration.

6. Methodology

The project follows these stages:

1. Import Python libraries.
2. Create the sample student dataset.
3. Export the original dataset into CSV format.
4. Load the CSV file using Pandas.
5. Inspect the initial data types and missing values.
6. Identify duplicate records.
7. Remove duplicate records.

8. Convert numerical columns into suitable formats.
9. Convert the date column into datetime format.
10. Calculate missing-value counts and percentages.
11. Apply median imputation to numerical columns.
12. Replace the missing categorical value with an Unknown category.
13. Verify the cleaned dataset.
14. Compare original and cleaned results.
15. Export the final cleaned dataset and summary.

7. Duplicate-Record Removal

Duplicate records represent repeated observations in a dataset.

The Pandas `duplicated()` function was used to identify repeated rows.

The `drop_duplicates()` function was then applied to remove duplicate records while keeping the first occurrence.

The results were:

- Original records: 12
- Duplicate records removed: 2
- Records after duplicate removal: 10

The cleaned dataset was verified to ensure that no exact duplicate rows remained.

8. Data-Type Correction

The original dataset contained numerical information stored as strings and an incorrectly formatted age entry.

The following conversions were performed:

Age

The age column was converted into an integer using `pd.to_numeric()`.

The incorrectly formatted value `twenty_two` was converted into a missing value using `errors="coerce"` and later replaced with the median age.

Marks

The marks column was converted into numerical values. Missing marks were subsequently replaced with the median.

Attendance

The attendance column was converted into numerical values and missing entries were filled using the median.

Joining Date

The joining-date column was converted into datetime format using `pd.to_datetime()`.

Department

The department column was retained as categorical information and its missing entry was replaced with `Unknown`.

Employee-style identifiers are not used in this dataset; student IDs and names remain strings.

9. Missing-Value Analysis

After duplicate removal and numerical conversion, the dataset contained four missing cells.

Column	Missing Values
department	1
age	1
marks	1
attendance	1

The missing numerical values were treated using median imputation.

The missing department was replaced with `Unknown`.

After treatment, the dataset contained zero missing values.

10. Missing-Value Treatment

Median Imputation

Median imputation was applied to age, marks, and attendance.

The median was selected because it is less affected by extreme values than the mean.

Categorical Treatment

The department column contained one missing value. It was replaced with the category `Unknown` to avoid assigning an unsupported department to that student.

These operations preserved all 10 records after duplicate removal.

11. Before-and-After Comparison

Metric	Before Cleaning / Stage	After Cleaning
Original records	12	10
Duplicate records removed	—	2
Missing cells after type conversion	4	0
Remaining duplicate records	—	0
Final records	—	10

The final dataset contains 10 student records with no missing values or exact duplicate rows.

12. Results and Observations

The data-cleaning process produced the following observations:

1. The original dataset contained 12 records.
2. Two exact duplicate records were identified and removed.
3. An incorrectly formatted age value was detected during numerical conversion.
4. Four missing cells were identified after type correction.
5. Numerical missing values were handled using median imputation.
6. The missing department was replaced with `Unknown`.
7. All final data types were corrected.
8. The cleaned dataset contains 10 records and zero missing values or exact duplicate rows.

13. Importance of Data Cleaning

Data cleaning improves the consistency and reliability of information.

Removing duplicates prevents repeated observations from influencing analysis.
Handling missing values helps ensure that important calculations can be performed.
Correcting data types ensures that numerical and date-based operations work correctly.

These techniques are important before conducting exploratory data analysis or training machine-learning models.

14. Applications

Basic data cleaning is useful in:

- Student performance analysis.
- Employee information management.
- Sales and financial reporting.
- Customer databases.
- Healthcare information systems.
- Business analytics.
- Machine-learning data preparation.

15. Limitations

The project uses a small, manually created dataset and demonstrates fundamental data-cleaning methods.

Median imputation is a simple treatment method and may not be suitable for every real-world dataset. Similarly, replacing missing categorical values with `Unknown` may require further consideration in more complex analyses.

16. Future Enhancements

The project could be extended by:

- Using larger real-world datasets.
- Comparing mean and median imputation.
- Handling inconsistent text values.
- Detecting near-duplicate records.
- Visualizing missing values.
- Developing reusable data-cleaning functions.
- Building an automated data-quality reporting workflow.

17. Conclusion

This project demonstrates how Python and Pandas can combine missing-value handling, duplicate removal, and data-type correction into a single data-cleaning workflow.

The original dataset contained 12 records, including two duplicate entries and four missing cells after type conversion. After cleaning, the final dataset contains 10 records with no missing values or exact duplicate rows.

The assignment provides practical experience in improving data quality before analysis and machine-learning applications.

18. References

1. Pandas Documentation: https://pandas.pydata.org/docs/
2. Pandas Missing Data Guide: https://pandas.pydata.org/docs/user_guide/missing_data.html

3. Pandas `drop_duplicates()` Reference: https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.drop_duplicates.html
4. Pandas `to_numeric()` Reference: https://pandas.pydata.org/docs/reference/api/pandas.to_numeric.html

19. Project Links

GitHub Repository: https://github.com/venky1318/Basic_Data_Cleaning